MATH 110 — Week 1

Notes Chapters 1–2; Homework 1

§1 Sets and functions

The question. Let $A = \{1, 2, 3, 4, 5, 6\}$ and $E = \{x \in A : x \text{ is even}\}$. How do we prove $E = \{2, 4, 6\}$? Check that each even element of A is on the list, and that each listed element is even. For a function, the related task is to determine which inputs and outputs a condition selects.

Definition (Notes Definition 1 and Example 1)**.** A *set* is a collection of distinguishable elements. Write $x \in A$ for membership and $B \subseteq A$ when every element of B belongs to A. The empty set $\varnothing$ has no elements. Relative to an ambient set S, define

$$A^c = \{x \in S : x \notin A\}, \quad A \cup B = \{x : x \in A \text{ or } x \in B\},$$

$$A \cap B = \{x : x \in A \text{ and } x \in B\}, \quad A \setminus B = A \cap B^c.$$

The Cartesian product $A \times B$ consists of ordered pairs (a, b) with $a \in A, b \in B$. Thus $(1, 2) \neq (2, 1)$. In Notes Example 1, $\{x \in \{1, 2, 3, 4, 5, 6\} : x \text{ is even}\} = \{2, 4, 6\}$; order and repetitions do not change a set.

Definition (Notes Definition 2)**.** A function $f : A \to B$ assigns exactly one value $f(a) \in B$ to every $a \in A$. It is *injective* if $f(a) = f(a')$ implies $a = a'$, *surjective* if every $b \in B$ equals some $f(a)$, and *bijective* if both hold. For $U \subseteq A$ and $V \subseteq B$, its image and inverse image are

$$f(U) = \{f(u) : u \in U\}, \qquad f^{-1}(V) = \{a \in A : f(a) \in V\}.$$

Here $f^{-1}(V)$ is defined even when f is not bijective.

Example (Notes Example 2)

Take $\mathbb{N} = \{1, 2, 3, \ldots\}$. The map $f(n) = n + 1$ is injective: $m + 1 = n + 1$ implies $m = n$. It is not surjective because 1 has no preimage.

Proposition (Notes Chapter 1, finite-set assertion)

If finite sets A, B have the same number n of elements, a function $f : A \to B$ is injective if and only if it is surjective.

Where the proof comes from

Want. Show that attaining all n outputs and giving different outputs to different inputs are equivalent.

Know. Listing the inputs as $a_1, \ldots, a_n$ lists every possible output as $f(a_1), \ldots, f(a_n)$, possibly with repetitions.

Key step. Count distinct values in this list. If none repeat, there are n values, enough to exhaust B. If two repeat, there are at most $n - 1$, too few to exhaust B.

Example. Three inputs sent to p, q, q attain only two outputs; a three-element codomain must have a missing element.

Proof. 1. If $n > 0$ and f is injective, $f(a_1), \ldots, f(a_n)$ are n distinct elements of B. Because B has exactly n elements, every element occurs: f is surjective.
2. If f is not injective, at least two entries of this n-entry list coincide. It has at most $n-1$ distinct values, so misses an element of B. Thus surjectivity implies injectivity as well.
3. If $n = 0$, both sets are empty. The empty function has no pair of distinct inputs and no missing codomain element, so it has both properties.

□

Proving a set identity by membership To prove $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$, test an arbitrary element x:

1. If $x \in A \cap (B \cup C)$, then $x \in A$ and either $x \in B$ or $x \in C$. In the first case $x \in A \cap B$; in the second $x \in A \cap C$. Both cases put x in the right-hand union.
2. Conversely, an element of the right-hand union is in $A \cap B$ or $A \cap C$. Either case gives $x \in A$ and $x \in B \cup C$, so $x \in A \cap (B \cup C)$.

The two implications prove both inclusions, hence equality.

Computing images and inverse images by tracing a function Let $X = \{a, b, c, d\}$, $Y = \{p, q, r\}$, with

$$f(a) = p, \qquad f(b) = p, \qquad f(c) = q, \qquad f(d) = q.$$

1. For $U = \{a, c\}$, use only arrows starting in U. Their endpoints give $f(U) = \{p, q\}$.
2. For $V = \{q, r\}$, collect *all* inputs whose arrows end in V. These are c, d, so $f^{-1}(V) = \{c, d\}$. No input maps to r.
3. Sending these inputs forward gives $f(f^{-1}(V)) = \{q\}$, which misses $r \in V$.
4. Starting with U instead, $f^{-1}(f(U)) = f^{-1}(\{p, q\}) = X$. It also contains b, d, because they share outputs with a, c.

Shortcuts

- To disprove injectivity, give two different inputs with the same output; this negates its defining implication.
- To disprove surjectivity, give one codomain element with no preimage; a missing output suffices.
- For $f(x) = x^2$ on $\mathbb{R}$, solve $f(x) \in [1, 4]$ as $1 \le x^2 \le 4$. The answer is $[-2, -1] \cup [1, 2]$; both signs are needed.

Traps

- Equal images need not mean equal inputs: for $f(x) = x^2$ on $\mathbb{R}$, $f(1) = f(-1)$.
- The codomain matters: $x \mapsto x^2$ from $\mathbb{R}$ to $\mathbb{R}$ is not onto, but the same rule with codomain $[0, \infty)$ is onto.

Do we recover the original selected set after both operations?

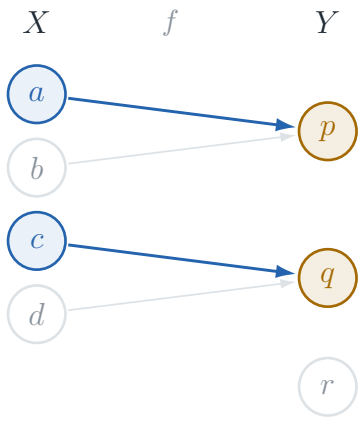
$f : X \to Y, \qquad X = \{a, b, c, d\}, \qquad Y = \{p, q, r\}$

All arrows show f: a, b go to p; c, d go to q; no input goes to r.

Starting with inputs: collecting all starting points can add inputs.

1 Follow the chosen inputs.

Choose $U = \{a, c\} \subseteq X$.

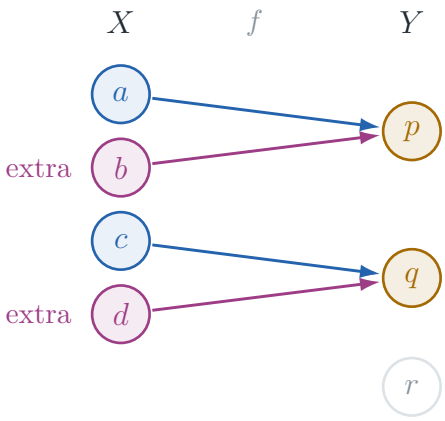


$f(U) = \{p, q\}$.

Only arrows from a, c select outputs.

2 Collect all starting points.

Keep the outputs $f(U) = \{p, q\}$.



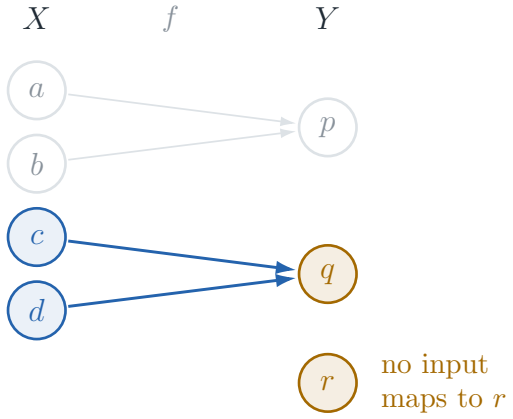
$f^{-1}(f(U)) = \{a, b, c, d\} \supsetneq U$.

b, d: extra inputs with the same outputs.

Starting with outputs: sending the collected inputs can lose outputs.

3 Collect all starting points.

Choose $V = \{q, r\} \subseteq Y$.

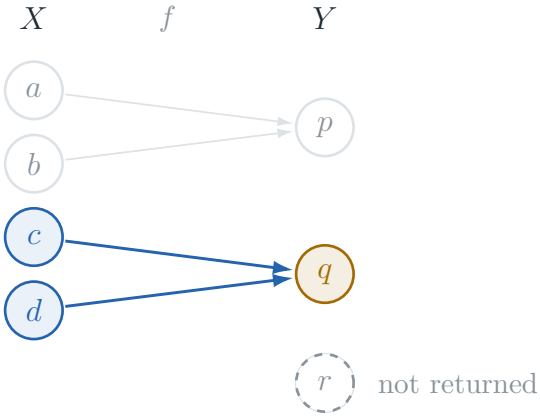


$f^{-1}(V) = \{c, d\}$.

Collect every input reaching q or r.

4 Send those inputs forward.

Keep the inputs $f^{-1}(V) = \{c, d\}$.



$f(f^{-1}(V)) = \{q\} \subsetneq \{q, r\} = V$.

q returns; r is not reached by any input.

§2 Fields and complex arithmetic

The question. Solve $2x = 1$ using an allowed number system. There is no integer solution, but $x = 1/2$ works in $\mathbb{Q}$. Which laws guarantee that $ax = b$ can be solved whenever $a \neq 0$?

Definition (Notes Definition 3)**.** A field F has addition and multiplication, distinct elements $0, 1$, and these laws: both operations are associative and commutative; $a + 0 = a$ and $1a = a$; $a(b + c) = ab + ac$; every a has an additive inverse $-a$; every nonzero a has a multiplicative inverse a^{-1}. Subtraction means addition of the additive inverse; division by $a \neq 0$ means multiplication by a^{-1}.

Proposition (Notes Chapter 1, identities, inverses and cancellation)

Identities and inverses are unique. In a field, $a + b = a + c$ implies $b = c$, $a0 = 0$, and $ab = 0$ implies $a = 0$ or $b = 0$.

Where the proof comes from

Want. Remove a common term from $a + b = a + c$, and a nonzero factor from $ab = 0$.

Know. The axioms provide $-a$ and, when $a \neq 0$, a^{-1}. Associativity lets us regroup $(-a)+(a+b)$ as $((-a)+a)+b = b$.

First try. Multiplying $ab = 0$ by a^{-1} is unavailable when $a = 0$; that case must be settled before using the inverse.

Key step. Add $-a$ to both sides for additive cancellation. To show $a0 = 0$, first use distributivity to obtain $a0 = a0 + a0$, then cancel $a0$. This supplies the zero-product calculation needed after multiplying by a^{-1}.

Example. Over $\mathbb{Q}$ or $\mathbb{R}$, in $3 + b = 3 + c$, adding -3 gives $b = c$. In $3b = 0$, multiplying by $1/3$ gives $b = 0$.

Proof. 1. Two additive identities satisfy $0 = 0 + 0' = 0'$. Two multiplicative identities satisfy $1 = 1 \cdot 1' = 1'$.

2. If b, c are additive inverses of a, then $b = b + (a + c) = (b + a) + c = c$.
3. If b, c are multiplicative inverses of a, then $ab = ac = 1$ and $b = b \cdot 1 = b(ac) = (ba)c = 1c = c$, by associativity and commutativity.
4. Add $-a$ to $a + b = a + c$ to obtain $b = c$ by associativity and the identity law.
5. Distributivity gives $a0 = a(0 + 0) = a0 + a0$. Comparing $a0 + 0 = a0 + a0$ and cancelling the common first term gives $0 = a0$.
6. If $ab = 0$ and $a \neq 0$, multiply by a^{-1} to obtain $b = a^{-1}0 = 0$.

□

Solving $ax = b$ in a field When $a \neq 0$, multiply by a^{-1} to isolate x; then check that the resulting value really solves the equation.

1. Any solution must satisfy $x = 1x = (a^{-1}a)x = a^{-1}(ax) = a^{-1}b$, using associativity and $ax = b$. Thus there is at most one solution.
2. The candidate does work: $a(a^{-1}b) = (aa^{-1})b = 1b = b$. Hence $x = a^{-1}b$ is the unique solution when $a \neq 0$.
3. If $a = 0$, then $ax = 0$ for every x. Thus $0x = 0$ has every $x \in F$ as a solution, while $0x = b$ has no solution when $b \neq 0$.
4. Over $\mathbb{R}$, $2x = 1$ gives $x = 2^{-1} \cdot 1 = 1/2$, and substitution gives $2(1/2) = 1$.

Example (Notes Example 3)

$\mathbb{Q}$ and $\mathbb{R}$ are fields with their usual arithmetic. In $\mathbb{Q}$, a nonzero p/q has inverse q/p. The integers are not a field: $2n = 1$ has no integer solution.

Definition (Notes Example 4; complex numbers)**.** A complex number has form $a+bi$ with real a, b and $i^2=-1$. Equality means equality of the real and imaginary parts. Identify a real number a with $a+0i$. Addition is componentwise, and

$$(a+bi)(c+di)=(ac-bd)+(ad+bc)i.$$

The conjugate of $z=a+bi$ is $\overline{z}=a-bi$.

Proposition (Notes Proposition 3; complex field laws: LADR 1.3)

For complex w, z,

$$\overline{w+z}=\overline{w}+\overline{z}, \quad \overline{wz}=\overline{w}\,\overline{z}, \quad z=\overline{z}\iff z\in\mathbb{R}, \quad z\overline{z}\geq 0.$$

The last quantity is real; it is zero only when $z=0$. For $z\neq 0$, $z^{-1}=\overline{z}/(z\overline{z})$. In particular $\mathbb{C}$ is a field.

Where the proof comes from

Want. Find a number $c+di$ such that $(a+bi)(c+di)=1$ when $(a,b)\neq(0,0)$, and check how conjugation interacts with arithmetic.

Know. The product has real part $ac-bd$ and imaginary part $ad+bc$. To equal 1, these must be 1 and 0 respectively.

First try. If $a\neq 0$, choosing $(c,d)=(1/a,0)$ gives $(a+bi)(1/a)=1+(b/a)i$, which is not 1 when $b\neq 0$. If $a=0$, $1/a$ is unavailable.

Key step. The choice $(c,d)=(a,-b)$ makes $ad+bc=-ab+ba=0$. Its real part is $a^2+b^2>0$, so divide both coordinates by that real number. The same sign change defines conjugation; expand each coordinate to check its sum and product laws.

Example. $(3+4i)(3-4i)=9-12i+12i-16i^2=25$, so $(3-4i)/25$ is the inverse of $3+4i$.

Proof. 1. Write $w=a+bi$, $z=c+di$. Conjugating their sum gives $(a+c)-(b+d)i=\overline{w}+\overline{z}$.

2. Conjugating their product gives $(ac-bd)-(ad+bc)i=(a-bi)(c-di)$.
3. $z=\overline{z}$ is equivalent to $2di=0$, or $d=0$.
4. $z\overline{z}=c^2+d^2\geq 0$, and it is positive if either real coordinate is nonzero.
5. Thus $z\,[\overline{z}/(c^2+d^2)]=1$ for $z\neq 0$. The additive inverse is $-c-di$.
6. Multiplication is commutative: $ac-bd=ca-db$ and $ad+bc=cb+da$. For distributivity, multiply $a+bi$ by $(c+e)+(d+f)i$: its coordinates are

$$(ac-bd)+(ae-bf), \qquad (ad+bc)+(af+be),$$

which are the sums of the coordinates of $(a+bi)(c+di)$ and $(a+bi)(e+fi)$.

7. For associativity, put $w=a+bi$, $z=c+di$, $t=e+fi$. Both $(wz)t$ and $w(zt)$ have real part $ace-adf-bcf-bde$ and imaginary part $acf+ade+bce-bdf$, by the multiplication formula.
8. Addition is associative because each real coordinate adds associatively. The real numbers $0, 1$ act as identities. Hence all field axioms hold.

□

Computing powers, products and reciprocals: Notes Example 4

1. Square before computing a cube:

$$(1+i)^2=1+2i+i^2=2i, \qquad (1+i)^3=2i(1+i)=2i+2i^2=-2+2i.$$

2. Distribute every term, then collect the two coordinates:

$$(5+i)(4+3i)=20+15i+4i+3i^2=(20-3)+(15+4)i=17+19i.$$

3. Multiply numerator and denominator by the denominator's conjugate:

$$\frac{1}{3+4i}=\frac{3-4i}{(3+4i)(3-4i)}=\frac{3-4i}{9+16}=\frac{3}{25}-\frac{4}{25}i.$$

Find w such that $(3+4i)w = 1$.

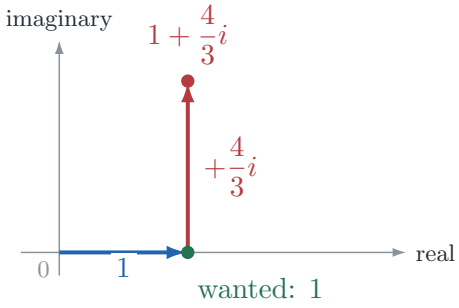
For $w = c + di$, $\quad (3+4i)w = \underbrace{(3c-4d)}_{\text{must be } 1} + \underbrace{(3d+4c)}_{\text{must be } 0} i.$

1 Try $w = 1/3$: the imaginary part remains.

$$w = \frac{1}{3}$$

$$(3+4i)\frac{1}{3} = 1 + \frac{4}{3}i$$

The real part is 1, but $4/3 \neq 0$.



2 Choose c, d so $3d + 4c = 0$.

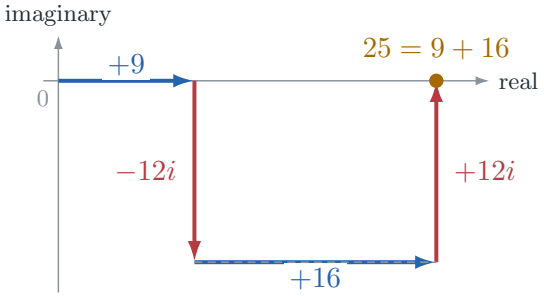
Set $c = 3$: $3d + 12 = 0$, so $d = -4$.

$w = 3 - 4i = \overline{(3+4i)}$

Multiply each row term by each column term:

	3	$-4i$
3	9	$-12i$
$4i$	$+12i$	$-16i^2 = +16$

$3(-4) + 4(3) = -12 + 12 = 0.$



Equal vertical steps cancel.
The product lies on the real axis.

3 The product is 25, so divide the candidate by 25.

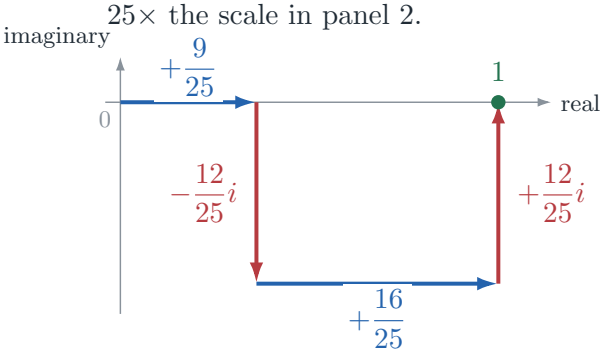
$$w = \frac{3-4i}{25} = \frac{3}{25} - \frac{4}{25}i$$

Every contribution to the product is now divided by 25.

$$\frac{9}{25} + \frac{16}{25} = 1$$

$$-\frac{12}{25} + \frac{12}{25} = 0$$

25× the scale in panel 2.



$$(3+4i)\left(\frac{3}{25} - \frac{4}{25}i\right) = \frac{9+16}{25} + \frac{-12+12}{25}i = 1.$$

A real answer obtained from complex intermediate values: the notes' cubic

1. Compute $(2+i)^2 = 4+4i+i^2 = 3+4i$, then
$$(2+i)^3 = (3+4i)(2+i) = 6+3i+8i+4i^2 = 2+11i.$$
2. Likewise $(2-i)^2 = 3-4i$ and $(2-i)^3 = (3-4i)(2-i) = 6-3i-8i+4i^2 = 2-11i$.
3. Thus $2+i$ is one cube root of $2+11i$, and $2-i$ is one cube root of $2-11i$. These choices add to 4; direct substitution gives $4^3 - 15\cdot 4 - 4 = 64 - 60 - 4 = 0$.

This verifies a solution of $x^3 - 15x - 4 = 0$ using the indicated choices of cube roots. It does not say that a complex number has a unique cube root.

Example (Notes Chapter 1, the field $\mathbb{F}_2$)

$\mathbb{F}_2 = \{0,1\}$ has addition and multiplication modulo 2:

$+$	0	1
0	0	1
1	1	0

$\cdot$	0	1
0	0	0
1	0	1

.

1. Two integers represent the same element when they differ by an even integer. Replacing a by $a+2k$ changes $a+b$ by $2k$ and ab by $2kb$, both even. Thus taking remainders gives well-defined addition and multiplication.
2. Taking the remainder of $(a+b)+c = a+(b+c)$ and $a(b+c) = ab+ac$ proves associativity of addition and distributivity; the integer commutative laws and multiplicative associative law give the corresponding identities in the same way.
3. The tables show that 0 is the additive identity and 1 is the multiplicative identity. Also $0+0 = 1+1 = 0$, so each element is its own additive inverse. The only nonzero element, 1, has inverse 1. These verify the remaining field axioms.

A polynomial expression in x is a finite sum of terms $a_k x^k$ with field coefficients, added and multiplied by distributivity and $x^j x^k = x^{j+k}$. The letter x is formal; no numerical value has to be substituted.

Shortcuts

- Compute z^{-1} by conjugating and dividing by the sum of squares of its real coordinates; their product is real and nonzero.
- Powers of i repeat every four exponents: $i^0 = 1$, $i^1 = i$, $i^2 = -1$, $i^3 = -i$, $i^4 = 1$. Thus $i^{11} = i^{4\cdot 2+3} = (i^4)^2 i^3 = -i$.
- In $\mathbb{F}_2$, $(x+y)^2 = x^2+xy+yx+y^2 = x^2+y^2$, because $xy = yx$ and $1+1 = 0$. For example, $(x+1)^2 = x^2+1$.

Traps

- $\mathbb{Z}$ is not a field even though addition and multiplication stay in $\mathbb{Z}$: $1/2$ is missing.
- In $\mathbb{F}_2$, cancellation of a nonzero factor works, but division by 2 does not: $2 = 0$ in this field.
- Taking real parts need not preserve multiplication. For $w = z = i$, the product $wz = -1$ has real part -1, while the product of the real parts is $0\cdot 0 = 0$.

§3 Vector spaces

The question. For tuples, $2(1,3)+(4,-1)=(6,5)$. For real functions $f(x)=x$ and $g(x)=1$, the function $2f+g$ has value $2x+1$ at x. Which laws let us calculate with these two kinds of objects in the same way?

Definition (Notes Definition 4; notes cite LADR 1.20)**.** A vector space V over a field F is a set with addition $V\times V\to V$ and scalar multiplication $F\times V\to V$. Addition is commutative and associative, has a zero vector 0_V, and every vector has an additive inverse. For $a,b\in F$ and $u,v\in V$,

$$(ab)v=a(bv),\quad a(u+v)=au+av,\quad (a+b)v=av+bv,\quad 1v=v.$$

An element of V is a vector; an element of F is a scalar. We write 0 for either zero when its meaning is clear. Vector subtraction is $u-v=u+(-v)$.

Proposition (Notes Chapter 2; LADR 1.26–1.27 as cited there)

The zero vector and each additive inverse are unique, and vector addition permits cancellation.

Where the proof comes from

Want. If two vectors both act as zero, show that they are equal; if two vectors both add to v to give zero, show they are equal.

Know. We can add and regroup vectors using the axioms, but we have not yet justified subtraction as cancellation.

Key step. Add the two candidate zero vectors together: each identity law identifies this sum with the other candidate. For inverses u,w, insert $v+w=0$ into $u+0$, then regroup to use $u+v=0$.

Example. The inverse of $(2,-3)$ must be $(-2,3)$: if $(2,-3)+(a,b)=(0,0)$, the coordinate equations force $a=-2, b=3$.

Proof. 1. For two zero vectors $0,0'$, $0=0+0'=0'$.

2. If $v+u=v+w=0$, then $u=u+(v+w)=(u+v)+w=w$.

3. If $v+u=v+w$, adding $-v$ to both sides gives $u=w$.

□

Example (Notes Example 5: tuples)

In F^n, addition and scaling are componentwise:

$$(x_1,\dots,x_n)+(y_1,\dots,y_n)=(x_1+y_1,\dots,x_n+y_n),\quad a(x_1,\dots,x_n)=(ax_1,\dots,ax_n).$$

1. Each coordinate stays in F, so both operations stay in F^n. The zero tuple is $(0,\dots,0)$ and the inverse of x is $(-x_1,\dots,-x_n)$.
2. To check $a(x+y)=ax+ay$, coordinate j on the left is $a(x_j+y_j)=ax_j+ay_j$, the coordinate on the right. The field distributive law proves equality in every coordinate.
3. Commutativity and associativity of addition, $(ab)x=a(bx)$, $(a+b)x=ax+bx$, and $1x=x$ follow respectively from those field laws in each coordinate. Hence F^n is a vector space. The case $n=1$ is F itself.

Example (Notes Example 6: functions and sequences)

For a set S, F^S consists of all functions $S \to F$. Define $(f+g)(s) = f(s) + g(s)$ and $(af)(s) = af(s)$. For $f(x) = x$, $g(x) = 1$ on $\mathbb{R}$, $(2f+g)(3) = 2f(3) + g(3) = 7$; the function itself is $x \mapsto 2x + 1$.

1. The zero function has value 0 at every input; the additive inverse $-f$ has value $-f(s)$ at input s.
2. For example, at an arbitrary s,

$$\big(a(f+g)\big)(s) = a\big(f(s) + g(s)\big) = af(s) + ag(s) = (af + ag)(s).$$

   The middle equality is distributivity in F. Equality at every input is equality of functions.
3. The other axioms use exactly the corresponding field identities at each input: for example $((ab)f)(s) = (ab)f(s) = a(bf(s)) = (a(bf))(s)$. Thus F^S is a vector space.
4. For $S = \{1, \ldots, n\}$, record f as the tuple $(f(1), \ldots, f(n))$. For $S = \mathbb{N} = \{1, 2, \ldots\}$, record the sequence $(f(1), f(2), \ldots)$; this space is also written F^∞.

Example (Notes Example 7: choosing a smaller scalar field)

$\mathbb{C}$ is a real vector space with its usual addition and multiplication by real scalars. For instance $2(1+i) - (3-i) = -1 + 3i \in \mathbb{C}$.

1. Real scaling and addition stay in $\mathbb{C}$. Its zero and additive inverses are the complex ones. Each required identity is a complex field identity applied to real scalars.
2. The same reasoning makes both $\mathbb{R}$ and $\mathbb{C}$ vector spaces over $\mathbb{Q}$: a rational scalar times an element stays in the set, and the field identities still hold.

Changing the scalar field changes the closure question. For example, $\mathbb{R}$ with its usual multiplication cannot be a vector space over $\mathbb{C}$, since $i \cdot 1 \notin \mathbb{R}$.

Proposition (Notes Proposition 4; notes cite LADR 1.30–1.32)

For $a \in F$ and $v \in V$, $0v = 0_V$, $a0_V = 0_V$, and $(-1)v = -v$.

Where the proof comes from

Want. Identify $0_F v$ and $a0_V$, which are not specified directly by the axioms, then identify $(-1)v$.

Know. If a vector q satisfies $q = q + q$, comparison with $q = q + 0_V$ and additive cancellation forces $q = 0_V$.

Key step. Produce that equation by writing $0_F = 0_F + 0_F$ in $0_F v$, and $0_V = 0_V + 0_V$ in $a0_V$. For $(-1)v$, add v: distributivity gives $(-1)v + v = (-1+1)v = 0_F v$.

Example. For $v = (2, -3)$, the resulting identities are $0_F v = (0, 0)$ and $(-1)v = (-2, 3)$.

Proof. 1. $0v = (0+0)v = 0v + 0v$; cancelling $0v$ gives $0v = 0_V$.

2. $a0_V = a(0_V + 0_V) = a0_V + a0_V$; cancelling $a0_V$ gives $a0_V = 0_V$.

3. $(-1)v + v = (-1+1)v = 0v = 0_V$. Uniqueness of the additive inverse gives $(-1)v = -v$.

□

Checking a proposed scalar multiplication On the usual additive set $\mathbb{R}$, define $a * v = 0$ for all a, v. Most scaling identities hold because both sides are zero, but $1 * 1 = 0 \neq 1$, so it is not a vector space. For example, $a * (v + w) = 0 = 0 + 0 = a * v + a * w$ does hold, so checking distributivity alone would miss the failure. The unit axiom must be checked as well.

Shortcuts

- For tuples, check an arbitrary coordinate: $[a(u+v)]_j = a(u_j+v_j) = au_j + av_j = [au+av]_j$. Since this holds for every j, the tuples agree.
- For functions, check an arbitrary input: $(f+g)(s) = f(s)+g(s) = g(s)+f(s) = (g+f)(s)$. Since this holds for every s, $f+g = g+f$.
- Check closure first. The positive real numbers fail under usual real scaling because $(-1)\cdot 1 = -1$ is outside the set; checking the remaining axioms cannot repair that failure.

Traps

- The scalar field cannot be omitted: $\mathbb{R} \subset \mathbb{C}$ is closed under real scaling but not complex scaling, since $i \cdot 1 = i \notin \mathbb{R}$.
- $0v = 0$ does not alone imply $v = 0$: any nonzero vector multiplied by the zero scalar is a counterexample.

§4 Subspaces, sums and direct sums

The question. In $\mathbb{R}^2$, let $U = \{(x,y) : x+y = 0\}$ and $W = \{(x,y) : x+y = 1\}$. Is each a vector space with the usual operations? The first task is to test $(0,0)$, then to substitute $au+v$ into each equation.

Definition (Notes Definition 5; notes cite LADR 1.34)**.** A subset U of a vector space V is a subspace if $0 \in U$ and $au+v \in U$ whenever $u, v \in U$, $a \in F$. Equivalently it contains zero and is closed under addition and scalar multiplication.

Proposition (Subspace test; Notes Definition 5 and its consequence)

The two formulations of the test agree. A subset satisfying the test is a vector space with the inherited operations.

Where the proof comes from

Want. Determine which vector-space axioms might fail when we keep only vectors in $U \subseteq V$.

Know. An identity such as $u+v = v+u$ is already true for elements of U, since it is true in V. The possible failures are that a result leaves U, that zero is missing, or that an additive inverse is missing.

Key step. Require $0 \in U$ and $au+v \in U$. Choosing $a = 1$ gives addition; choosing $v = 0$ gives scaling; choosing $a = -1$ then gives the needed additive inverses. Every operation and required element therefore remains available inside U.

Proof. 1. The combined condition with $a = 1$ gives addition; with $v = 0$ it gives scalar closure. Conversely the two closure properties give the combined condition.

2. $0 \in U$, and $-u = (-1)u \in U$ by scalar closure and Proposition 4.
3. The inherited associative, commutative, distributive and unit identities hold because they hold for these elements in V.

□

Substituting into a defining equation

1. The zero vector satisfies $x+y = 0$. For $u = (r,-r)$, $v = (s,-s)$ and $a \in \mathbb{R}$, $au+v = (ar+s, -ar-s)$; its coordinates add to 0. Thus U in the question box is a subspace.
2. The zero vector does not satisfy $x+y = 1$. Thus W is not a subspace. Also $(1,0) \in W$ but $2(1,0) = (2,0) \notin W$, a separate failure of scalar closure.

Example (Notes Example 8)

Continuity means that function values approach the value at each point as the input approaches that point; differentiability means that the difference quotient has a limit, denoted $f'(x)$. The closure and derivative rules below are cited from elementary calculus.

1. Within the space of real functions on $(0,1)$, continuous functions form a subspace: the zero function is continuous and if f, g are continuous and $a \in \mathbb{R}$, then $af + g$ is continuous.
2. Differentiable functions also form a subspace: the zero function is differentiable, and $(af+g)' = af' + g'$.

Example (LADR 1.35(d): a prescribed derivative)

For $b \in \mathbb{R}$, let D_b be the differentiable real-valued functions f on $(0,3)$ such that $f'(2) = b$. Here $f'(2)$ is the derivative of f at the input 2. Then D_b is a subspace of $\mathbb{R}^{(0,3)}$ if and only if $b = 0$.

1. If D_b is a subspace, it contains the zero function. Its derivative is zero at every input, so its membership forces $b = 0$.
2. Conversely, D_0 contains the zero function. For $f, g \in D_0$ and $a \in \mathbb{R}$, the cited derivative rules give that $af + g$ is differentiable and
$$(af+g)'(2) = af'(2) + g'(2) = a \cdot 0 + 0 = 0.$$
Thus $af + g \in D_0$, and the subspace test applies.
3. The condition does not require the function itself to be zero: $f(x) = (x-2)^2$ has $f'(x) = 2(x-2)$, so $f'(2) = 0$.

Example (LADR 1.35(e): complex sequences tending to zero)

A real sequence r_n *tends to zero* if, for every real $\varepsilon > 0$, there is an index N such that $-\varepsilon < r_n < \varepsilon$ whenever $n \geq N$. A complex sequence $z_n = x_n + iy_n$ tends to zero when both real sequences x_n and y_n tend to zero. We cite the elementary real-limit fact that sums and fixed real scalar multiples of sequences tending to zero still tend to zero.

1. The zero sequence tends to zero. Suppose $z_n = x_n + iy_n$ and $w_n = s_n + it_n$ tend to zero, and let $a + bi \in \mathbb{C}$. All four real-coordinate sequences tend to zero, and
$$(a+bi)z_n + w_n = (ax_n - by_n + s_n) + i(ay_n + bx_n + t_n).$$
2. Each displayed real-coordinate sequence tends to zero by the cited real-limit fact. Thus the complex sequence $(a+bi)z + w$ tends to zero. This proves that these sequences form a subspace of $\mathbb{C}^\infty$.

Example (Notes Example 9)

In $\mathbb{R}^{\mathbb{N}_0}$, where $\mathbb{N}_0 = \{0, 1, 2, \ldots\}$, the sequences satisfying $x_{n+2} = x_{n+1} + x_n$ form a subspace.

1. The zero sequence satisfies the condition. If x, y satisfy it, then
$$(ax+y)_{n+2} = a(x_{n+1} + x_n) + (y_{n+1} + y_n) = (ax+y)_{n+1} + (ax+y)_n.$$
2. Starting from $x_0 = p, x_1 = q$, the recurrence gives
$$x_2 = p + q, \quad x_3 = p + 2q, \quad x_4 = 2p + 3q, \quad x_5 = 3p + 5q.$$
With $(p, q) = (0, 1)$, this gives $(0, 1, 1, 2, 3, 5, 8, \ldots)$.
3. If two such sequences have the same first two entries, the formula makes their third entries equal; equal consecutive entries make the next entries equal. Induction shows equality at every index. Thus the first two entries determine the entire sequence.

Proposition (Notes Proposition 5; notes cite LADR 1C.10)

The intersection $U \cap W$ of two subspaces of V is a subspace.

Where the proof comes from

Want. Starting from u, v that lie in both U and W, show $au + v$ still lies in both.

Know. Closure in U proves membership in U only; it does not by itself prove membership in W.

Key step. Use $u, v \in U$ to obtain $au + v \in U$, and separately use $u, v \in W$ to obtain $au + v \in W$. These two conclusions are precisely membership in the intersection.

Example. For the two coordinate axes in $\mathbb{R}^2$, simultaneous membership requires both coordinates to be zero; their intersection is $\{(0, 0)\}$.

Proof. 1. Both subspaces contain 0, hence so does their intersection.
2. If $u, v \in U \cap W$, then $au + v$ lies in U and in W, hence in $U \cap W$.

□

Definition (Sum; Notes Proposition 6)**.** $U + W = \{u + w : u \in U, w \in W\}$. More generally $U_1 + \cdots + U_m$ consists of sums $u_1 + \cdots + u_m$ with $u_j \in U_j$. We abbreviate $u_1 + \cdots + u_m$ by $\sum_{j=1}^{m} u_j$, or $\sum_j u_j$ when the range is clear; $\sum_{j<k}$ means indices $1, \ldots, k-1$.

Proposition (Notes Proposition 6; notes cite LADR 1.40)

$U + W$ is a subspace and is the smallest subspace containing $U \cup W$.

Where the proof comes from

Want. A subspace containing both U and W, with no unnecessary vectors.

First try. Use $U \cup W$. For the coordinate axes, this contains $(1, 0)$ and $(0, 1)$ but omits $(1, 1)$, so it is not closed under addition.

Know. Any subspace containing U and W must contain every $u + w$. This forces the entire set $U + W$ to be included.

Key step. Check that the set of all $u+w$ is itself closed: regroup $a(u_1+w_1)+(u_2+w_2)$ as $(au_1+u_2)+(aw_1+w_2)$. Its first part is still in U and its second still in W, so we have not needed any vectors outside $U + W$.

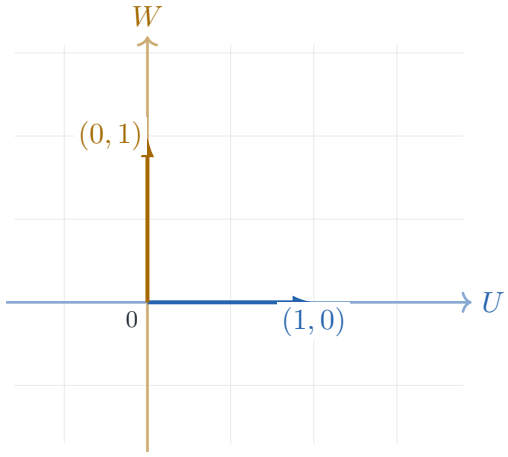
Proof. 1. $0 = 0 + 0$ lies in $U + W$. Also $a(u_1 + w_1) + (u_2 + w_2) = (au_1 + u_2) + (aw_1 + w_2) \in U + W$.
2. Each $u \in U$ equals $u + 0$ and each $w \in W$ equals $0 + w$, so $U \cup W \subseteq U + W$.
3. Any subspace X containing $U \cup W$ contains every $u + w$ by addition closure; hence $U + W \subseteq X$.
4. For m subspaces, $0 = 0+\cdots+0$ is in the sum. Also $a(u_1+\cdots+u_m)+(v_1+\cdots+v_m) = (au_1+v_1)+\cdots+(au_m+v_m)$, with $au_j + v_j \in U_j$ by closure in U_j. Thus the same test proves $U_1 + \cdots + U_m$ is a subspace.

□

Why do we need the sum instead of the union?

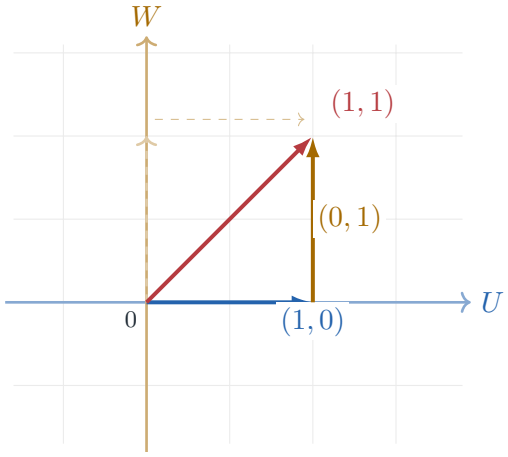
$U = \{(x, 0) : x \in \mathbb{R}\}$ $W = \{(0, y) : y \in \mathbb{R}\}$

1 Choose two vectors in the union.



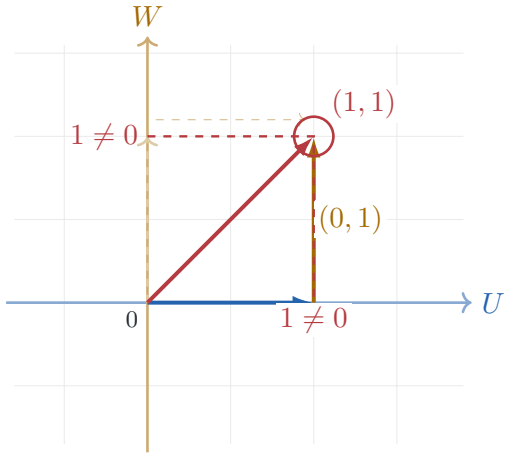
The union contains vectors on either axis.
Both chosen vectors belong to $U \cup W$.

2 Add them: translate the gold arrow.



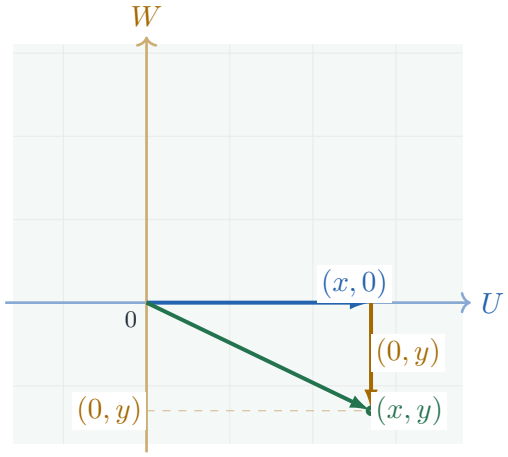
Translation keeps the gold vector $(0, 1)$.
The endpoint is $(1, 0) + (0, 1) = (1, 1)$.

3 The sum is outside the union.



Both coordinates are nonzero.
The sum lies on neither axis, so
$U \cup W$ is not closed under addition.

4 Allow sums: reach any target.



$(x, y) = (x, 0) + (0, y)$ for any real x, y.
Every target is a sum: $U + W = \mathbb{R}^2$.

Computing a sum from coordinate conditions: LADR Example 1.38 and equation 1.39 Let

$$U = \{(a, a, b, b) : a, b \in F\}, \qquad W = \{(c, c, c, d) : c, d \in F\}.$$

Both are subspaces of F^4: zero satisfies their coordinate equalities, and, for $\alpha \in F$, $\alpha u + v$ preserves any equality $u_j = u_k$, $v_j = v_k$, because $\alpha u_j + v_j = \alpha u_k + v_k$.

To find $U + W$, first add arbitrary summands to find a necessary coordinate condition. Then construct summands for every vector satisfying that condition; the first calculation alone gives only one inclusion.

1. A general sum is
$$(a, a, b, b) + (c, c, c, d) = (a + c, a + c, b + c, b + d).$$
Its first two coordinates are equal. Hence, as in LADR equation 1.39,
$$U + W \subseteq \{(x, x, y, z) : x, y, z \in F\}.$$

2. For the reverse inclusion, start with any (x, x, y, z). Choose $(x, x, y, y) \in U$ to match its first three coordinates. The remaining difference is $(0, 0, 0, z - y)$, which belongs to W because its first three coordinates are equal. Thus
$$(x, x, y, z) = (x, x, y, y) + (0, 0, 0, z - y) \in U + W.$$

3. Both inclusions give $U + W = \{(x, x, y, z) : x, y, z \in F\}$. The third and fourth coordinates are unrestricted; their equality in U need not persist after adding an element of W.

Definition (Notes Definition 6; two-subspace criterion: LADR 1.46)**.** The sum of U, W is *direct*, written $U \oplus W$, when $U \cap W = \{0\}$. A sum of m subspaces is direct when

$$U_k \cap (U_1 + \cdots + U_{k-1}) = \{0\} \qquad (2 \le k \le m).$$

Proposition (Notes Proposition 7; notes cite LADR 1.41, 1.45)

For subspaces $U_1, \ldots, U_m$, the following are equivalent: (i) their sum is direct; (ii) every vector in their sum has a unique representation $u_1 + \cdots + u_m$, $u_j \in U_j$; (iii) $u_1 + \cdots + u_m = 0$ implies $u_1 = \cdots = u_m = 0$.

Where the proof comes from

Want. Show that two expressions $v = u_1 + \cdots + u_m = v_1 + \cdots + v_m$ must have the same summands exactly when the sum is direct.

Know. Subtracting the expressions gives a sum equal to zero. If the last summands differ, their difference belongs both to U_m and to the sum of the preceding subspaces:

$$u_m - v_m = (v_1 - u_1) + \cdots + (v_{m-1} - u_{m-1}).$$

Key step. Directness makes this difference zero. To avoid assuming all earlier summands already agree, choose the *largest* index where they differ; the identical later terms cancel. The same intersection condition forces that last difference to be zero, a contradiction.

Example. For two subspaces, $u + w = u' + w'$ gives $u - u' = w' - w \in U \cap W$. If $x \in U \cap W$ were nonzero, then $(u + x) + (w - x) = u + w$ would give different summands from the same two subspaces.

Plan. Prove directness gives uniqueness, uniqueness gives only the zero representation of zero, and failure of directness produces a nonzero summand in a representation of zero.

Proof. 1. Assume (i). Each vector in the sum has at least one representation by the definition of sum. Suppose two representations differ, and let k be the largest index where $u_k \neq v_k$.

2. Cancel the terms for $j > k$. If $k = 1$, the remaining equation gives $u_1 = v_1$, a contradiction. If $k \geq 2$, rearrange it to obtain
$$u_k - v_k = \sum_{j<k}(v_j - u_j) \in U_k \cap (U_1 + \cdots + U_{k-1}) = \{0\}.$$
Each difference on the right lies in its indicated subspace by closure. Again $u_k = v_k$, a contradiction. This proves (ii).

3. Assume (ii). The vector 0 has the representation $0 + \cdots + 0$. Uniqueness says that any $u_1 + \cdots + u_m = 0$ has each $u_j = 0$, which is (iii).

4. Assume (iii). If (i) fails, some $k \geq 2$ admits $0 \neq x \in U_k \cap (U_1 + \cdots + U_{k-1})$. Write $x = u_1 + \cdots + u_{k-1}$ with $u_j \in U_j$.

5. Then $u_1 + \cdots + u_{k-1} + (-x) + 0 + \cdots + 0 = 0$ has its kth summand $-x \neq 0$ in U_k, contradicting (iii). Thus (i) holds.

□

Finding summands by solving their coordinate equations Let $U = \{(t, t) : t \in \mathbb{R}\}$ and $W = \{(s, -s) : s \in \mathbb{R}\}$. Both are subspaces: zero has parameter zero, and $a(t, t) + (r, r) = (at + r, at + r)$, with the corresponding calculation for W.

1. To write (x, y) as a sum, solve
$$(x, y) = (t, t) + (s, -s) = (t + s, t - s), \qquad t + s = x, \quad t - s = y.$$

2. Adding the equations gives $2t = x + y$; subtracting the second from the first gives $2s = x - y$. Thus
$$(x, y) = \left(\frac{x+y}{2}, \frac{x+y}{2}\right) + \left(\frac{x-y}{2}, -\frac{x-y}{2}\right).$$
These values exist and are forced, so $\mathbb{R}^2 = U \oplus W$ by Proposition 7.

3. The intersection test gives the same conclusion: $(t, t) = (s, -s)$ requires $t = s$ and $t = -s$. Hence $2t = 0$, so $t = s = 0$ over $\mathbb{R}$.

4. For $(x, y) = (4, 2)$, $2t = 6$ and $2s = 2$, so $(4, 2) = (3, 3) + (1, -1)$.

Find the two summands of $v = (4, 2)$.

$U = \{(t, t) : t \in \mathbb{R}\}$ $\quad$ $W = \{(s, -s) : s \in \mathbb{R}\}$

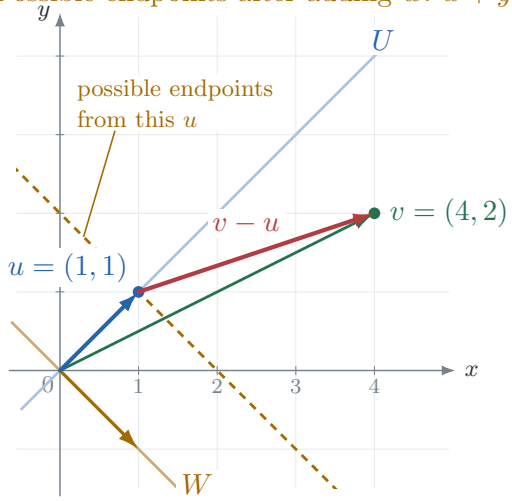
Every $w = (s, -s) \in W$ has coordinate sum $s + (-s) = 0$.

From $u = (t, t)$, every possible endpoint $u + w$ has $x + y = 2t$.

This line passes through u and is parallel to W.

1 Try $u = (1, 1)$.

Possible endpoints after adding w: $x + y = 2$

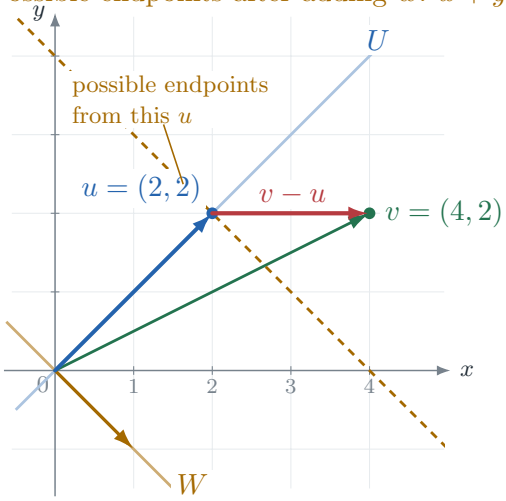


$v - u = (4, 2) - (1, 1) = (3, 1)$.

$3 + 1 \neq 0$, so $(3, 1) \notin W$.

2 Move to $u = (2, 2)$.

Possible endpoints after adding w: $x + y = 4$

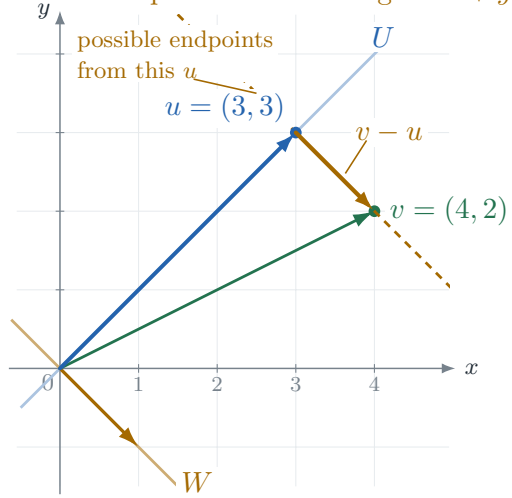


$v - u = (4, 2) - (2, 2) = (2, 0)$.

$2 + 0 \neq 0$, so $(2, 0) \notin W$.

3 Move to $u = (3, 3)$.

Possible endpoints after adding w: $x + y = 6$

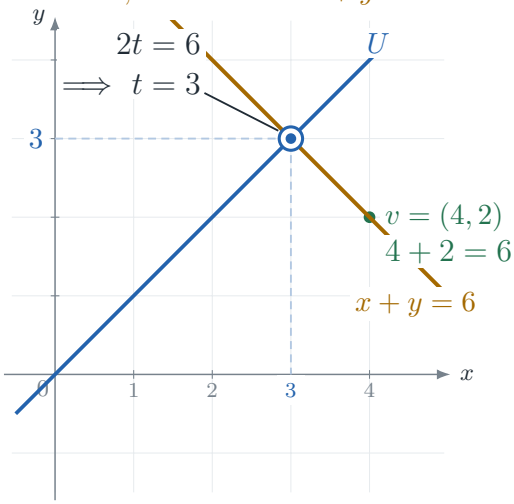


$v - u = (4, 2) - (3, 3) = (1, -1) \in W$.

$(4, 2) = (3, 3) + (1, -1)$.

4 Why is this the only choice?

To reach v, u must have $x + y = 6$.



The two lines meet only once: $u = (3, 3)$.

$t + s = 4, \quad t - s = 2 \implies t = 3, \ s = 1.$

Why pairwise intersections do not decide a sum of three subspaces Set $U_1 = \{(t, 0)\}$, $U_2 = \{(0, t)\}$, $U_3 = \{(t, t)\}$, with all parameters real. Each is a subspace by the same parameter calculation used above.

1. A point on two of these distinct lines has both the relevant zero coordinate and the equation $x = y$ when applicable; in every pair this forces $(x, y) = (0, 0)$.
2. Nevertheless $(1, 1)$ has representations

$$(1, 0) + (0, 1) + (0, 0) = (0, 0) + (0, 0) + (1, 1).$$

   The summands in each expression belong in order to U_1, U_2, U_3. They differ, so the sum is not direct.
3. Specifically, $U_1 + U_2 = \mathbb{R}^2$, hence $U_3 \cap (U_1 + U_2) = U_3 \neq \{0\}$. This is the intersection required in Definition 6; pairwise intersections do not check it.
4. Also $U_2 + U_3 = \mathbb{R}^2$, since $(x, y) = (0, y - x) + (x, x)$. Therefore

$$U_1 \cap (U_2 + U_3) = U_1, \qquad (U_1 \cap U_2) + (U_1 \cap U_3) = \{0\}.$$

   For example $(1, 0)$ belongs to the left side but not the right. Intersection does not distribute over subspace sum.

Shortcuts

- To disprove the subspace property, first test zero. The set $x + y = 1$ fails immediately.
- A homogeneous equation $c_1x_1 + \cdots + c_nx_n = 0$ (right-hand side zero) with fixed field coefficients passes the test: zero satisfies it and $\sum_j c_j(au_j + v_j) = a\sum_j c_ju_j + \sum_j c_jv_j = a0 + 0 = 0$. For example, $x - 2y = 0$ is preserved because $a(2r, r) + (2s, s) = (2(ar + s), ar + s)$.
- For two subspaces, solve $u = w$ to test their intersection, as in $(t, t) = (s, -s)$ above. For more than two, solve $u_1 + \cdots + u_m = 0$ and use Proposition 7; the three-line example shows why pairwise tests can miss a nonzero solution.

Traps

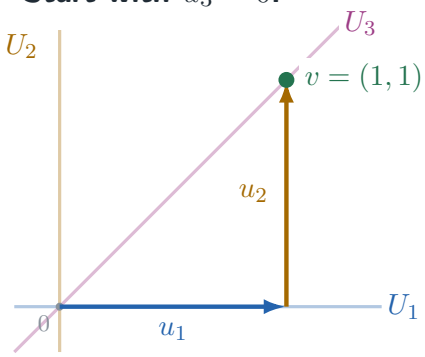
- A union need not be a subspace: the two coordinate axes contain $(1, 0)$ and $(0, 1)$ but not their sum $(1, 1)$.
- Let $U_1 = \{(t, 0)\}$, $U_2 = \{(0, t)\}$, $U_3 = \{(t, t)\}$ (all parameters real). Their pairwise intersections are zero, but $(1, 0) + (0, 1) + (-1, -1) = 0$, so their sum is not direct.
- For these same lines, $U_1 \cap (U_2 + U_3) = U_1$, while $(U_1 \cap U_2) + (U_1 \cap U_3) = \{0\}$. Intersection does not distribute over subspace sum.

Do three different lines give unique summands?

$U_1 = \{(x, 0)\}$, $U_2 = \{(0, y)\}$, $U_3 = \{(t, t)\}$ over $\mathbb{R}$.

Each pair meets only at 0. Keep the target $v = (1, 1)$ fixed.

1 Start with $u_3 = 0$.



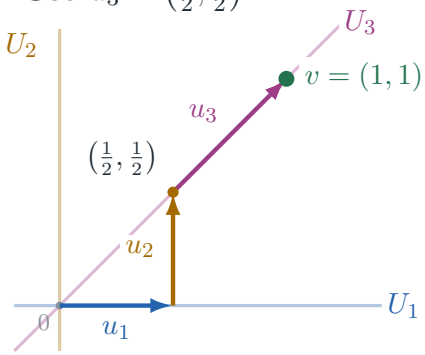
$u_1 = (1, 0)$

$u_2 = (0, 1)$

$u_3 = (0, 0)$

The two axis vectors already add to v.

2 Use $u_3 = \left(\frac{1}{2}, \frac{1}{2}\right)$.



$u_1 = \left(\frac{1}{2}, 0\right)$

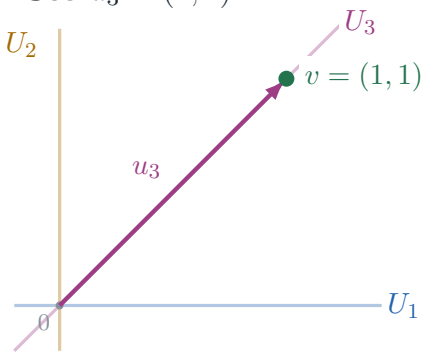
$u_2 = \left(0, \frac{1}{2}\right)$

$u_3 = \left(\frac{1}{2}, \frac{1}{2}\right)$

The axis vectors add to $\left(\frac{1}{2}, \frac{1}{2}\right)$.

Adding the diagonal vector still gives v.

3 Use $u_3 = (1, 1)$.



$u_1 = (0, 0)$

$u_2 = (0, 0)$

$u_3 = (1, 1)$

The diagonal vector alone equals v.

The summands changed; v did not.

The same vector has different summands. This sum is not direct.

§5 Problem set

HW1 P1. For $f : X \to Y$, $U \subseteq X$, $V \subseteq Y$, prove (a) $f(f^{-1}(V)) \subseteq V$, (b) $U \subseteq f^{-1}(f(U))$. (c) Give strict-inclusion examples and determine exactly when equality holds in each.
Hint: Use the image and inverse-image definitions; distinguish a fixed subset from a statement for every subset.

HW1 P2 (LADR 1A.7). Show that $(-1 + \sqrt{3}\,i)/2$ is a cube root of 1.
Hint: Expand using $i^2 = -1$ and square before multiplying once more.

HW1 P3. For $a, b \in F$, prove $ab = 0$ implies $a = 0$ or $b = 0$.
Hint: Apply the multiplicative-inverse axiom only in the case $a \neq 0$.

HW1 P4 (LADR 1B.2). If $a \in F$, $v \in V$ and $av = 0$, prove $a = 0$ or $v = 0$.
Hint: Use scalar associativity and Notes Proposition 4.

HW1 P5 (LADR 1C.4). For $b \in \mathbb{R}$, let U_b consist of continuous real-valued functions f on $[0, 1]$ satisfying $\int_0^1 f(x)\,dx = b$. Prove that U_b is a subspace of $\mathbb{R}^{[0,1]}$ if and only if $b = 0$.
Hint: Test the zero function, then use the subspace test and linearity of the integral (from calculus).

HW1 P6 (LADR 1C.5). Is $\mathbb{R}^2$ a subspace of the complex vector space $\mathbb{C}^2$?
Hint: The scalar field is $\mathbb{C}$; test multiplication by i.

HW1 P7. In $\mathbb{F}_2 = \mathbb{Z}/2\mathbb{Z}$, expand $(x + 1)^8$ by first computing $(x + 1)^2$ and repeatedly squaring. Explain how this differs from the calculation over $\mathbb{R}$.
Hint: In $\mathbb{F}_2$, the coefficient 2 is zero.

Exam practice (materials/exams/midterm1-review.pdf, Section II, Exercise 7). Prove that the intersection of subspaces is a subspace. What about union?
Hint: Apply Notes Proposition 5, and test the two coordinate axes for union.